Describing Welfare

Session 2

Ethan Ligon

September 10, 2026

Introduction

Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates*
    - Deaton & Zaidi §§2–4 if you read nothing else
- **Christiaensen, Ligon, & Sohnesen (2022)**: at least memorize the title.

Measuring Individual Welfare

Many notions of individual welfare; generally lead to some unidimensional index, say w (bigger is better):

- Subjective ("How happy are you?")
- Utility (How is this different from Subjective happiness?)
- Material (Something about command over economic resources)
    - Wealth?
    - Income?
    - Consumption expenditures?

Aside: What about risk?

Material Well-Being: Statistics

Some summary statistic of the *distribution* F of w in the population.

Inequality E.g., Atkinson's (1971) utilitarian notion of inequality:

$$I = U\left(\int_{\underline{w}}^{\bar{w}} w dF(w)\right) - \int_{\underline{w}}^{\bar{w}} U(w) dF(w) \geq 0$$

Poverty A quantile of some function $\phi_z(w)$; e.g.,

$$P(z) = \int_{\underline{w}}^{z} \phi_z(w) dF(w)$$

These are distributions *of the population*, so in practice those sample weights will matter!

Conventional options for w

Income

- Difficult to measure if sources are agricultural or informal
- Also highly variable—a farmer's main income may arrive once a year, after harvest, and be very risky.

Wealth

- What is it?
- Perhaps something like discounted expected income?
- But if income is difficult to measure then all possible future incomes may be much more difficult

Conventional options for w, continued

Assets

- Related to wealth, but may be measurable now
- Financial assets + Bicycles + Goats + House = ?? In principle these could be priced, but this may not be at all practical.

Consumption Expenditures
The "Gold standard"

- Probably easier to measure than income
- Consumption is smoother than income
- And closer to the theoretical object (permanent income, lifetime resources) that we really want.

The Theoretical Object

The household's problem

What is the consumption aggregate a measure *of*? Assume:

(A1) Preferences $U(c)$ over J goods, increasing, strictly concave, the same every period, discounted at β.

(A2) One decision maker the household maximizes a single U.

(A3) One asset a risk-free bond, gross return R, no borrowing limit.

(A4) Expectations $\mathbb{E}_t$ over the true conditional distribution.

With A_t resources on hand and $x_t = p_t \cdot c_t$ total expenditure,

$$V_t(A_t, p_t) = \max_{c_t} U(c_t) + \beta \mathbb{E}_t V_{t+1}(A_{t+1}, p_{t+1}),$$

subject to $A_{t+1} = R(A_t - x_t) + y_{t+1}$.

The marginal utility of expenditure

Differentiate. With $u_j = \partial U / \partial c_j$, the first-order condition in c_j and the envelope condition in A are

$$u_j(c_t) = \lambda_t \, p_{tj}, \qquad \lambda_t \equiv \beta R \, \mathbb{E}_t[\partial V_{t+1} / \partial A] = \partial V_t / \partial A.$$

So λ_t is two things at once, which is what makes it useful:

- the multiplier on the period budget constraint, the utility value of one more cedi spent *today*;
- the derivative of the *value function*, the utility value of one more cedi of resources, the whole future included.

V is concave in A, so λ falls as the household gets richer. Write $w_t = -\log \lambda_t$, which therefore rises with welfare. This is another candidate for the w of the opening slide.

The martingale condition

Substituting the envelope condition into the definition of λ_t,

$$\lambda_t = \beta R \, \mathbb{E}_t \lambda_{t+1}.$$

When $\beta R = 1$ the marginal utility of expenditure is a martingale:

$$\mathbb{E}_t \lambda_{t+1} = \lambda_t.$$

A statement about information: whatever the household knew at t is already in λ_t, so nothing dated t forecasts the change.

- Needs A1–A4 *and* an interior solution: no binding borrowing limit.
- With several assets, $\beta \mathbb{E}_t[(\lambda_{t+1}/\lambda_t) R^j_{t+1}] = 1$ for every asset j held in positive quantity.

Why total expenditure?

Three steps, each with its own assumption. Let $v(x,p) = \max\{U(c) : p \cdot c \le x\}$, so $\lambda = \partial v/\partial x$.

1. At *given* prices x orders welfare, since $\lambda > 0$. Nothing beyond A1 is needed, but this ranks within a market only.
2. To rank across markets we need a scalar $P(p)$ with $v(x,p) = \psi(x/P(p))$. Such a P exists if and only if U is homothetic (A5), and then $P = e(p)$, the cost of a unit of the aggregate. Quasi-homothetic (A5′): two indices, a line $a(p)$ and a deflator $b(p)$.
3. So $\lambda = \psi'(x/e(p))/e(p)$, decreasing in real expenditure.

Under A5, real total expenditure $x/e(p)$ is a sufficient statistic for within-period welfare. It is what the rest of this session builds.

Hall's random walk

The strict permanent income hypothesis adds:

(A8) Constant prices or a martingale in p independent of λ, which puts the walk in quantities.

(A9) Quadratic utility $U(x) = -\frac{1}{2}(\bar{x} - x)^2$, so $\lambda_t = \bar{x} - x_t$ is *linear* in expenditure.

(A10) $\beta R = 1$ with R constant and known.

The martingale in λ is then a martingale in x:

$$\mathbb{E}_t[\bar{x} - x_{t+1}] = \bar{x} - x_t \quad \Longrightarrow \quad \mathbb{E}_t x_{t+1} = x_t.$$

Consumption is a random walk (Hall 1978): nothing dated t forecasts Δx_{t+1}, lagged income included. Testable, but it needs longitudinal data.

The assumptions, collected

	Assumption	What it buys
A1–A2	Time-separable U, one maximizer	a well-defined λ
A3	One asset, no borrowing limit	an interior solution
A4	Rational expectations	$\mathbb{E}_t$ is the truth
A5	Homotheticity, or quasi-	a deflator, or a line and one
A6	Iso-elastic cardinalization	w affine in log real x
A7	Durables weakly separable	nondurables and services
A8–A10	Constant p, quadratic U, $\beta R = 1$	Hall's random walk

Everything from here to the end of the session is an attempt to measure x. None of it repairs an assumption in this table.

Practical Difficulties in Measuring Individual Items of Consumption

The four components of consumption expenditures

Or: "Nondurable consumption and services".

Consumption expenditures in the LSMS surveys generally fall into one of five categories, and are typically elicited in different modules.

1. **Food** — purchased, own-produced, and received as gifts
2. **Non-food, non-durable** — fuel, transport, clothing, school fees, health
3. **Durables** — *value of services*, or "use value", not purchase price
4. **Housing** — rent, actual or imputed
5. **Other Services** — Not really measured at all

List Length

Beegle et al (2012): the *number* of goods asked has a large effect on recall, and on that evidence many NSOs have lengthened the list. Just for food:

Ethiopia	25 → 26 → 55 → 73 → 74
GhanaLSS	58 → 58 → 117 → 112 → 114 → 114 → 179
Burkina Faso	55 → 130 → 152
Uganda	61 → 63 → 70 → 71 → 77 → 77 → 117 → 116
Nigeria	90 → 90 → 96 → 104 → 116 → 116 → 110 → 110

- Should make measured total consumption more accurate;
- BUT: Really complicates comparisons across time.
- And: Can greatly increase cost of fielding the survey.

Timing & Recall

The model treats time as discrete. What in the field corresponds to a "period"? LSMS practice reveals what the designers thought.

Food Usually a seven-day recall, with interesting variation and some explicit experiments (Beegle et al. 2012) on how well respondents remember purchases.

Non-food expenditures or "semi-durables" Think things like clothing, inventories of kerosene.... Often a 30-day recall.

Durables Bicycles, radios, fans,.... Often a year-long recall for purchases, and/or an inventory of the *stock* of durables.

Food prices: which price?

Food *expenditures* are quantities times prices. Which prices?

- farmgate price received — what the household gave up
- local market price — what it would have cost
- these differ by the marketing margin, said to be 30–50%

Deaton & Zaidi: value at the price the household *faces*. In practice, a median unit value by item, cluster and round, falling back up the hierarchy when cells are thin.

Which of the two does a survey's own valuation question measure?

Food prices: what GLSS7 asks, and what it gets

Section 8 Part H, question 9: **"for how much would you sell one unit of … now?"** A farmgate question, by its wording.

Against what neighbours *paid*, same item, unit and cluster:

- Ghana 2016-17, 322 cells: median ratio **1.000**; a third tie exactly
- Uganda, eight waves, the same question: **1.000** pooled
- Uganda also records sales: the valuation is **1.23** times the sale price; buyers paid **1.60** times it

Asked what a bowl of maize would sell for, a household answers with the price of a bowl of maize. The margin is real — 1.3 to 2.2 where a survey records sales — and this question does not measure it.

Food quantities, and what was never bought

Much of the food in an LSMS household was never bought:

- Purchased at market
- Consumed out of own production
- Gift/Transfer
- Consumed outside the household (?)

Not a canard. In Ghana 2016-17 own production is **28.7%** of the value of food; 12,191 of 13,942 households record some. Deaton and Zaidi's Table 3.1 implies 31.9%.

But it reaches your aggregate only if you *value* it: from 1991-92 on the survey records own production in quantities, not cedis.

→ notebook

GLSS7's answer: a diary, visited six times

The most recent GhanaLSS is ambitious about frequent purchases: a diary, and an enumerator visiting six times at five-day intervals.

The form (`reading/`): six visit columns headed *2nd* to *7th*, asking **"…since my last visit?"**, and no price column anywhere.

Expensive; two experiments have asked whether it is worth it.

Scott & Amenuvegbe (1990) Ghana, and the reason GLSS asks this way

Beegle et al. (2012) Tanzania, eight modules, randomly assigned

The diary

CLUSTER NUMBER: ☐☐☐ HOUSEHOLD NUMBER: ☐☐

NAME OF HOUSEHOLD HEAD: .... DATE: ☐☐☐☐☐☐☐☐
DAY MTH YEAR

INSTRUCTION: RECORD ALL PAYMENTS MADE AND ESTIMATE ALL OWN PRODUCED GOODS CONSUMED.

Date	Place of Purchase/ Own Produce	Description of Item Purchased or Consumed	Qty	Unit	Price	Amount

"Record all payments made and *estimate all own produced goods consumed*": the diary asks for a price on own produce. Section 9B, filled from it, does not.

→ notebook

How fast does recall decay?

Scott & Amenuvegbe (1990), 13 frequently purchased items in Ghana:

- Reported daily expenditure falls steadily as the recall window lengthens from one day to one week, then flattens between one and two weeks
- Roughly **2.9% lost for every day added**, after reweighting away purchase cycles and start-up bias
- The decline is larger for items bought more often
- A one-year recall is simply erratic

So a weekly recall understates frequent purchases by around **20%** — and that was, in their words, "an abnormally well controlled field situation."

What actually drives the error

Beegle et al. (2012): eight modules randomly assigned across ~4,000 Tanzanian households, benchmarked against a personal diary.

- Recall *length* is the small knob: 7 vs 14 days moves food spending **7%**
- Item *list* design is the big one: **21%** between 58 items and 11 categories
- "Usual month" fails outright, and contaminates the module after it
- Household diaries miss about **19%** of what personal diaries catch

Same households: poverty from **47.5% to 66.8%**, by module alone.

Timing: what is a period?

For food, a day, probably: people schedule around meals. Surveys pick something longer, and each choice buys one error with another.

- A *diary* of what was eaten, meal by meal, is the ideal and the burden: hard to sustain, and paying for it buys a record, not an accurate one
- A *short* window is cheap but noisy: day-to-day variation in intake is not the error we are after
- *Usual* consumption asks the respondent to do our averaging for us, and to share our idea of "typical"; Beegle et al. found it fails outright

Inventories: purchases are not consumption

Storable staples — dry maize, rice, oil — are bought in bulk and drawn down. Perishables — fruit, fresh fish — are bought as eaten.

- A short window catches the fruit and misses the maize sack, or catches the sack and triples the week
- Averages out across households if purchase timing is unrelated to the visit; does not average out *within* one, which is where the line is applied
- Who buys in bulk is a matter of wealth: cash on hand, a dry store, a refrigerator. So the noise is on the rich, and on the staples
- Thirty days of visits (GLSS7) is one answer; asking what was *consumed*, as 8H does and 9B does not, is the other

Price Indices

Two different price indices, meant to address two different problems:

- **Temporal**: prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- **Spatial**: prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

A household-specific Paasche index

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j s_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with s_{ij} household i's budget share on item j.

$\rightarrow$ notebook

Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r,

$$\text{use value} \;\approx\; S\left(r + \frac{1}{L}\right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

Intra-Household Allocation

Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, usually conflated:

- **Household composition**: children cost less than adults — $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age,sex}}$
- **Economies of scale**: shared goods, so cost rises less than proportionately — $\Rightarrow n_i^\theta$, with $\theta \in (0, 1)$

Combined: $c_i = \dfrac{C_i}{A_i^\theta}$.

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

Barten: children re-price goods

A^θ makes a child a fraction of an adult *for everything*. Barten (1964): composition changes *which* goods a household needs,

$$\max_c \; U\left(\frac{c_1}{m_1(z)}, \ldots, \frac{c_J}{m_J(z)}\right) \quad \text{s.t.} \quad \sum_j p_j c_j = x,$$

so a household with demographics z is the reference household at prices $p_j m_j(z)$. A child makes milk dear; substitution does the rest.

- the scale is a ratio of cost functions, $e(u, p \circ m(z))/e(u, p)$: it depends on prices, and in general on u
- A^θ is the case where every m_j is the same number: children change the *size* of the bundle, never its *shape*
- a shape that depends on z is not what A5 allows

Constructing Aggregate Welfare Measures

Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z,

$$P_\alpha = \frac{1}{N} \sum_{i:c_i<z} \left(\frac{z - c_i}{z} \right)^\alpha .$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.
- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z.
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Weighted, always: $P_\alpha = \sum_i w_i \left(\frac{z-c_i}{z}\right)^\alpha / \sum_i w_i$.

$\rightarrow$ notebook

Inequality

- **Lorenz curve**: cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- **Gini**: twice the area between the Lorenz curve and the 45° line. Bounded, familiar, and insensitive where it matters most.
- **Generalized entropy** $GE(\alpha)$: decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

Inequality: Atkinson

The opening slide's $I = U\left(\int w\,dF\right) - \int U(w)\,dF$ is in utils, which nobody reports. Let w_{ede} solve $U(w_{\text{ede}}) = \int U\,dF$: the level which, given to everyone, yields the same total utility as the actual spread. Then $I = U(\bar{w}) - U(w_{\text{ede}})$, and

$$A = 1 - \frac{w_{\text{ede}}}{\bar{w}}$$

is the same judgement in cedis per cedi, with $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$ and ε assumed, like θ.

$A = 0.3$: this society would give up 30% of mean consumption to have the rest shared equally. Prefer the Lorenz curve to any scalar.

$\rightarrow$ notebook

One index, or several?

Everything here divides nominal expenditure by a scalar: a Paasche index over space, a CPI over time, an equivalence scale over composition. Three scalars, one household, one number out.

- one deflator is exact at **rank one**: budget shares do not vary with total expenditure
- but they do — Engel's law, which opened this session
- a line *and* a deflator are exact at **rank two** (Gorman), which Engel's law permits
- so on what grounds is a two-index procedure adequate?

Hold the question: session 3 answers it for Uganda.

Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent. Recompute P_0, P_1, P_2. Notebook: `poverty_line.ipynb`
2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north–south poverty gradient survives. Notebook: `spatial_deflation.ipynb`
3. Vary θ in $c_i = C_i/A_i^\theta$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change? Notebook: `equivalence_scales.ipynb`